

AI Sponsor

Building Tools with GenAI

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1. Problem Statement

Many personal health applications allow users to track alcohol consumption, but they typically provide only raw data such as drink counts, calories, or cost. This information is often presented without context, making it difficult for users to interpret their behavior or understand potential health implications. This creates a gap between data collection and meaningful insight. While users may consistently log their consumption, they are not given the tools to analyze patterns, assess risk, or make informed decisions. Generic health advice is widely available, but it is often not tailored to individual behavior, reducing its effectiveness. Additionally, many existing systems do not leverage modern AI techniques to enhance user understanding. Without contextual grounding, AI-generated feedback can be too general and lack relevance to the user's specific habits. The core problem is the absence of a system that can take structured user data and convert it into personalized, context-aware insights. This project addresses that need by combining data aggregation, contextual

knowledge retrieval, and AI-driven analysis to provide meaningful feedback that supports better decision-making.

2. Methodology and Implementation

The system is implemented as a cloud-based pipeline that transforms user data into AI-generated insights through a series of structured steps. The process begins with an HTTP-triggered Python cloud function, which receives a user ID (UID) from the client. The function validates the input and queries the Firestore database for all drink-related entries associated with that user. These entries include values such as drink count, calories, and cost. Rather than sending raw data to the AI model, the system aggregates the data into summary metrics. These totals are then converted into a natural language summary, which serves as input for the AI. This step simplifies the data and ensures compatibility with language model processing.

To improve the quality of the output, a Retrieval-Augmented Generation (RAG) approach is used. A predefined knowledge base containing health-related information is embedded into vector form. The user's summary is also embedded, and cosine similarity is used to identify the most relevant knowledge entries. The top results are selected as contextual guidance. The summary and retrieved context are combined into a structured prompt and sent to an OpenAI language model. The model generates a concise insight and recommendation based on both the user's data and the retrieved information.

3. Tools and Technologies

This project integrates several tools across frontend development, backend processing, and AI-based analysis. The frontend was built using FlutterFlow, which is a low-code/no-code platform for building cross-platform apps. The exported Flutter code allows the app to be run locally using the Flutter SDK. Firebase was used as the backend platform. Firestore serves as the database for storing user drink data, while Firebase Authentication provides user

management and unique user IDs (UIDs) for linking data to individuals. The core logic was implemented using a Python cloud function. This function handles incoming requests, retrieves and aggregates user data from Firestore, and prepares it for analysis.

The OpenAI API was used for both generating insights and creating embeddings. The language model produces natural language feedback, while the embedding model enables semantic comparison for the Retrieval-Augmented Generation (RAG) process. Cosine similarity was implemented in Python to compare embeddings and retrieve the most relevant contextual information from the knowledge base. Git and GitHub were used for version control and project management, ensuring the codebase is organized and reproducible. Overall, these tools work together to support data collection, processing, and AI-driven insight generation within the application.

4. Results

The project successfully demonstrates a working system that transforms user drink data into personalized, AI-generated insights. The application is able to retrieve user-specific data from Firestore,

aggregate it into meaningful metrics, and generate concise feedback using a Retrieval-Augmented Generation (RAG) approach. The integration of contextual knowledge improves the relevance of the AI output, resulting in insights that are more informative than generic responses. The system also delivers results in a clean, structured format suitable for real-time use within the mobile application. Overall, the project validates that combining data aggregation with AI and RAG can effectively bridge the gap between raw data and actionable user feedback.